# AstroPhot–GALFIT Validation with Physical Sigma Maps

FRB Host Inclination Analysis

March 18, 2026

## 1 Introduction

We performed AstroPhot and GALFIT analyses using survey-derived physical sigma maps (rather than flat or dummy variance assumptions), so the likelihood weighting is tied to measured noise structure in each image. For AstroPhot, we used PSF convolution and enforced a fixed Sersic index $n = 1$ for all FRB hosts to match the targeted modeling configuration. We then checked consistency against GALFIT (also from sigma-weighted PSF runs) and validated physical plausibility using non-parametric morphology from statmorph (CAS, Gini, M20), with explicit checks for extreme outliers.

## 2 AstroPhot (PSF + sigma, fixed $n = 1$) vs GALFIT

Table 1 reports side-by-side values for $R_e$, axis ratio $(b/a)$, and inclination angle $i$ for all FRBs. The objective is direct comparability of structural geometry between tools under physical-noise weighting.

Table 1: AstroPhot (PSF+sigma, fixed $n = 1$) vs GALFIT (PSF+sigma): side-by-side structural parameters.

| FRB | $R_{e,\mathrm{AP}}$ | $R_{e,\mathrm{GF}}$ | $(b/a)_{\mathrm{AP}}$ | $(b/a)_{\mathrm{GF}}$ | $i_{\mathrm{AP}}$ (deg) | $i_{\mathrm{GF}}$ (deg) |
|---|---|---|---|---|---|---|
| 20171020A | 15.51 | 45.89 | 0.364 | 0.390 | 68.65 | 70.02 |
| 20180924B | 0.64 | 2.40 | 0.634 | 0.650 | 50.63 | 50.86 |
| 20190102C | 0.97 | 3.77 | 0.379 | 0.400 | 67.70 | 69.30 |
| 20190608B | 1.34 | 4.94 | 0.689 | 0.680 | 46.43 | 48.45 |
| 20190714A | 1.05 | 4.15 | 0.346 | 0.380 | 69.74 | 70.75 |
| 20191001A | 1.68 | 6.61 | 0.520 | 0.520 | 58.69 | 60.67 |
| 20200906A | 1.61 | 6.21 | 0.310 | 0.320 | 71.97 | 75.23 |
| 20210320C | 0.94 | 3.61 | 0.683 | 0.700 | 46.90 | 46.79 |
| 20210410D | 0.87 | 2.99 | 0.383 | 0.140 | 67.50 | 90.00 |
| 20210807D | 3.88 | 13.37 | 0.715 | 0.700 | 44.39 | 46.79 |
| 20211127I | 6.14 | 15.50 | 0.953 | 0.910 | 17.55 | 25.03 |
| 20211203C | 0.72 | 2.76 | 0.394 | 0.420 | 66.81 | 67.86 |
| 20211212A | 3.20 | 13.01 | 0.827 | 0.830 | 34.26 | 34.70 |
| 20220207C | 7.43 | 29.79 | 0.208 | 0.210 | 77.98 | 86.25 |
| 20220307B | 1.48 | 6.30 | 0.628 | 0.650 | 51.09 | 50.86 |
| 20220310F | 1.12 | 4.30 | 0.682 | 0.690 | 47.02 | 47.62 |
| 20220319D | 1.08 | 3.24 | 0.895 | 0.900 | 26.50 | 26.42 |
| 20220509G | 3.37 | 11.12 | 0.393 | 0.400 | 66.83 | 69.30 |
| 20220825A | 1.56 | 5.89 | 0.629 | 0.630 | 51.05 | 52.43 |
| 20220912A | 1.27 | 5.42 | 0.616 | 0.640 | 51.95 | 51.65 |
| 20220914A | 1.17 | 3.92 | 0.541 | 0.490 | 57.28 | 62.84 |
| 20220920A | 1.84 | 6.87 | 0.863 | 0.860 | 30.29 | 31.39 |
| 20221012A | 1.31 | 4.98 | 0.545 | 0.550 | 56.96 | 58.47 |

Figure 1 shows the CDF bias comparison using the fixed-$n$ AstroPhot results against GALFIT and the SDSS-based reference envelope.

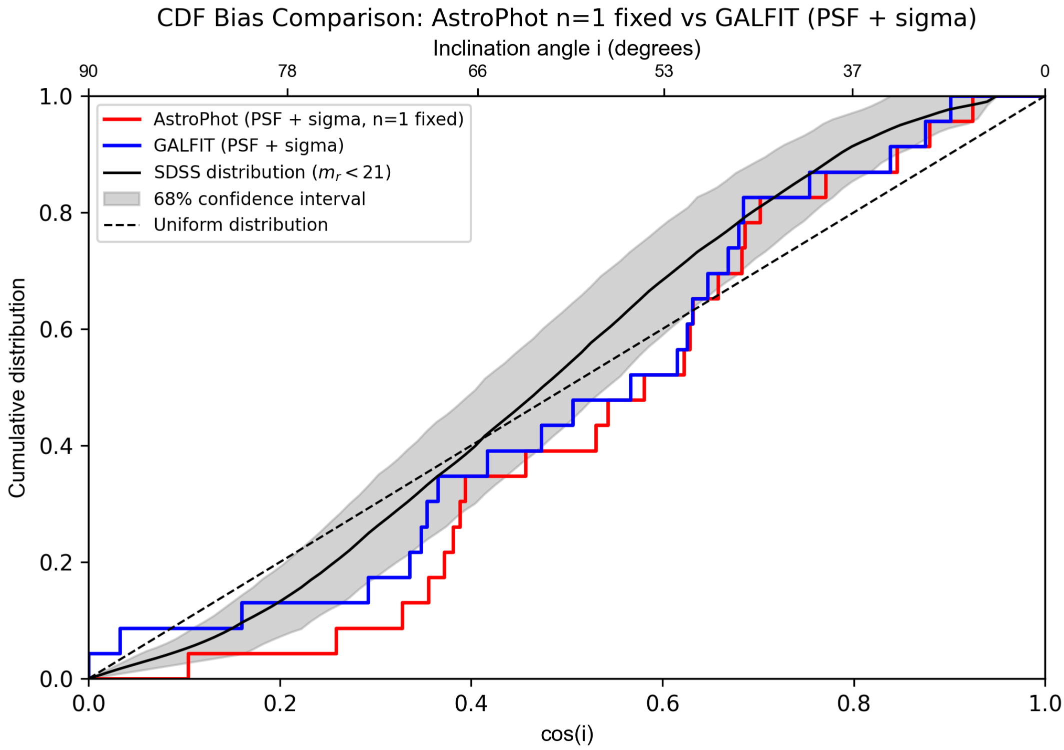


Figure 1: CDF bias comparison for AstroPhot (PSF + sigma, fixed $n = 1$) and GALFIT (PSF + sigma).

## 3 Statmorph Non-Parametric Validation (CAS, Gini, M20)

To ensure optimization remains physically grounded, we include non-parametric morphology measurements for all FRB hosts. Table 2 lists concentration ($C$), asymmetry ($A$), smoothness ($S$), Gini, and M20 from statmorph.

Table 2: Statmorph non-parametric morphology for all FRB hosts (CAS, Gini, M20).

| FRB | C | A | S | Gini | M20 |
|---|---|---|---|---|---|
| 20171020A | 2.159 | -0.026 | 0.002 | 0.420 | -1.202 |
| 20180924B | 2.834 | 0.006 | 0.057 | 0.531 | -1.723 |
| 20190102C | 2.172 | -0.141 | 0.020 | 0.395 | -1.540 |
| 20190608B | 2.993 | 0.015 | -0.003 | 0.544 | -1.842 |
| 20190714A | 2.286 | 0.003 | 0.000 | 0.423 | -1.491 |
| 20191001A | 2.235 | 0.135 | 0.041 | 0.429 | -1.536 |
| 20200906A | 2.483 | 0.146 | 0.075 | 0.429 | -1.438 |
| 20210320C | 2.439 | -0.038 | 0.042 | 0.441 | -1.693 |
| 20210410D | 2.305 | -0.088 | 0.000 | 0.473 | -1.597 |
| 20210807D | 2.553 | -0.210 | 0.012 | 0.467 | -1.802 |
| 20211127I | 2.692 | -0.262 | -0.042 | 0.521 | -1.938 |
| 20211203C | 2.724 | -0.195 | 0.000 | 0.468 | -1.811 |
| 20211212A | 2.466 | 0.166 | 0.004 | 0.406 | -1.688 |
| 20220207C | 2.402 | -0.125 | 0.009 | 0.416 | -1.325 |
| 20220307B | 2.182 | -0.201 | -0.013 | 0.379 | -1.410 |

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| FRB | C | A | S | Gini | M20 |
|---|---|---|---|---|---|
| 20220310F | 2.493 | -0.039 | 0.034 | 0.430 | -1.658 |
| 20220319D | 3.947 | -0.085 | -0.016 | 0.581 | -2.414 |
| 20220509G | 1.823 | 0.596 | 0.053 | 0.538 | -1.022 |
| 20220825A | 2.328 | -0.114 | 0.057 | 0.426 | -1.157 |
| 20220912A | 2.208 | -0.147 | 0.030 | 0.445 | -1.580 |
| 20220914A | 2.286 | 0.014 | 0.037 | 0.429 | -1.613 |
| 20220920A | 2.448 | 0.012 | -0.002 | 0.466 | -1.695 |
| 20221012A | 2.555 | -0.001 | -0.002 | 0.464 | -1.695 |

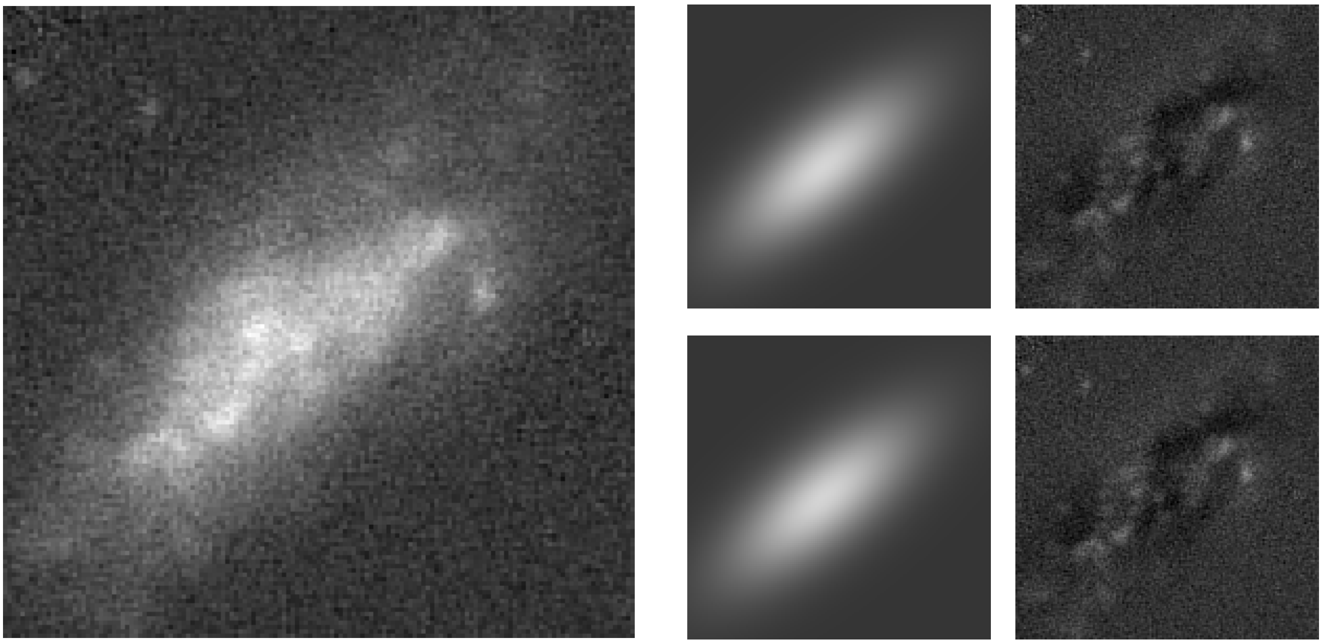
# 4 Conclusion

Using physical sigma maps in both fitting pipelines, AstroPhot (with PSF and fixed $n = 1$) and GALFIT show similar structural trends in $R_e$, $(b/a)$, and inclination angle. The CDF bias comparison supports population-level consistency, and the statmorph CAS/Gini/M20 check does not indicate catastrophic outlier behavior that would invalidate the fixed-$n$ optimization setup.

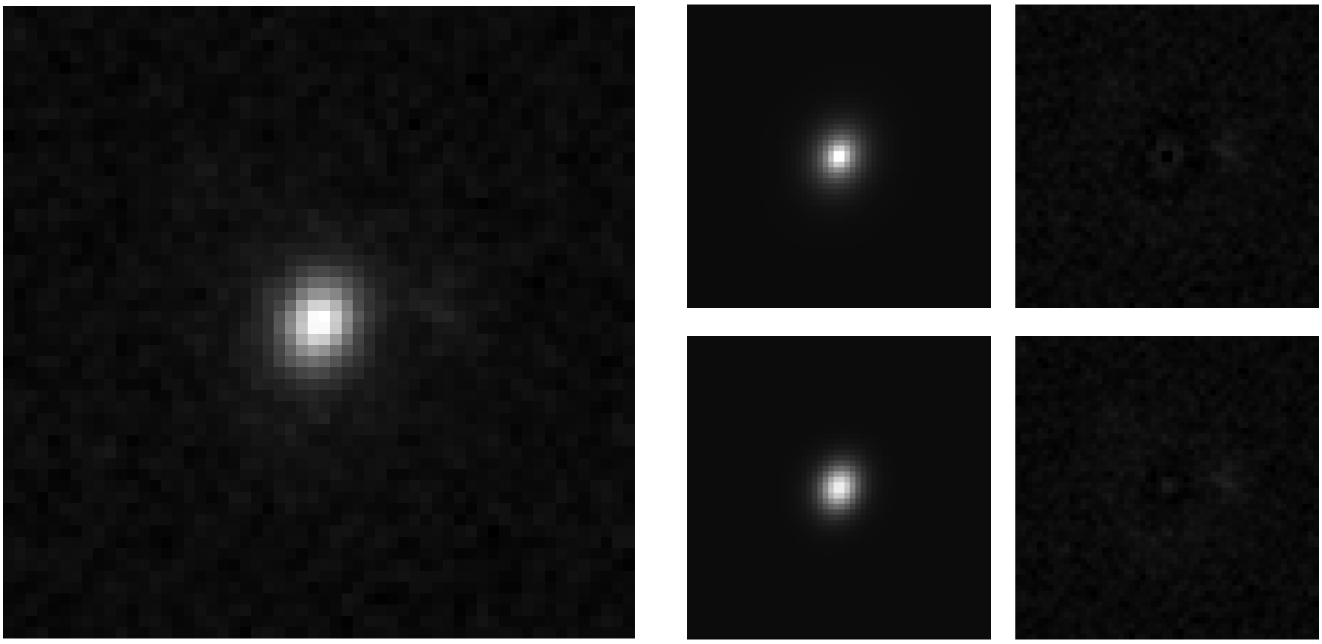
# 5 GALFIT Fitting Display (Sigma Runs)

This appendix follows the same visual format used in the reference LaTeX layout: one image panel and four model/residual panels for each FRB.
All panels use the same linear image scaling per FRB (image min/max reused for model and residual).

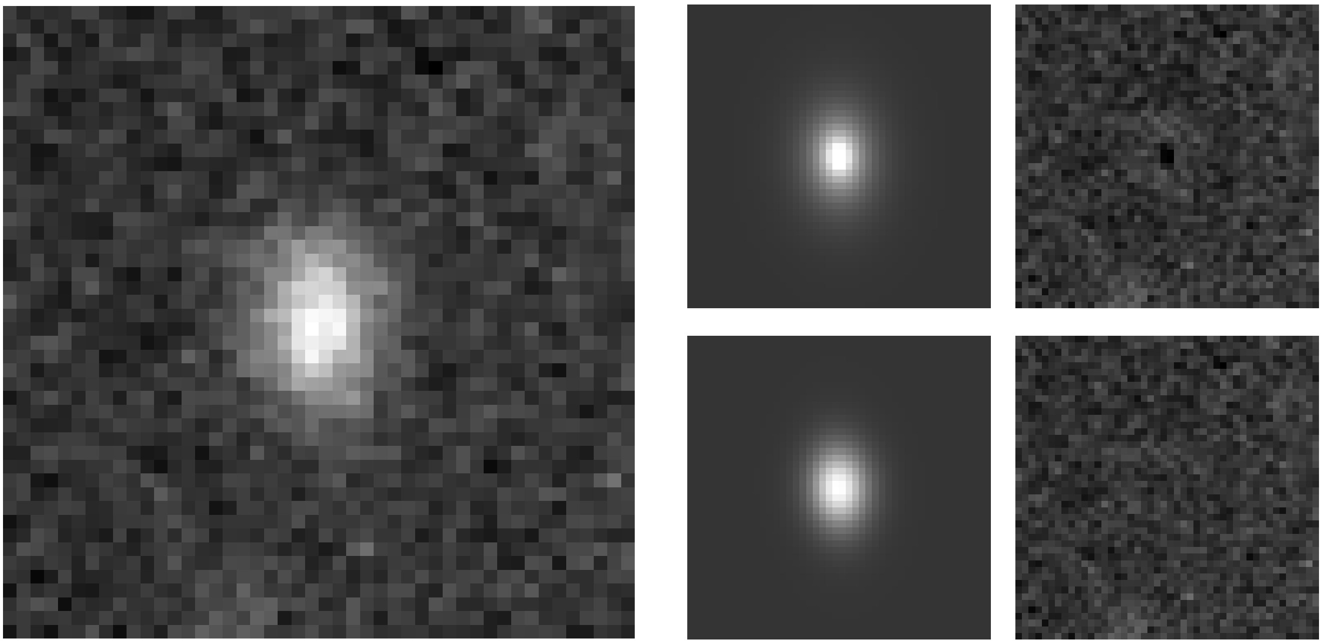
## 5.1 FRB 20171020A



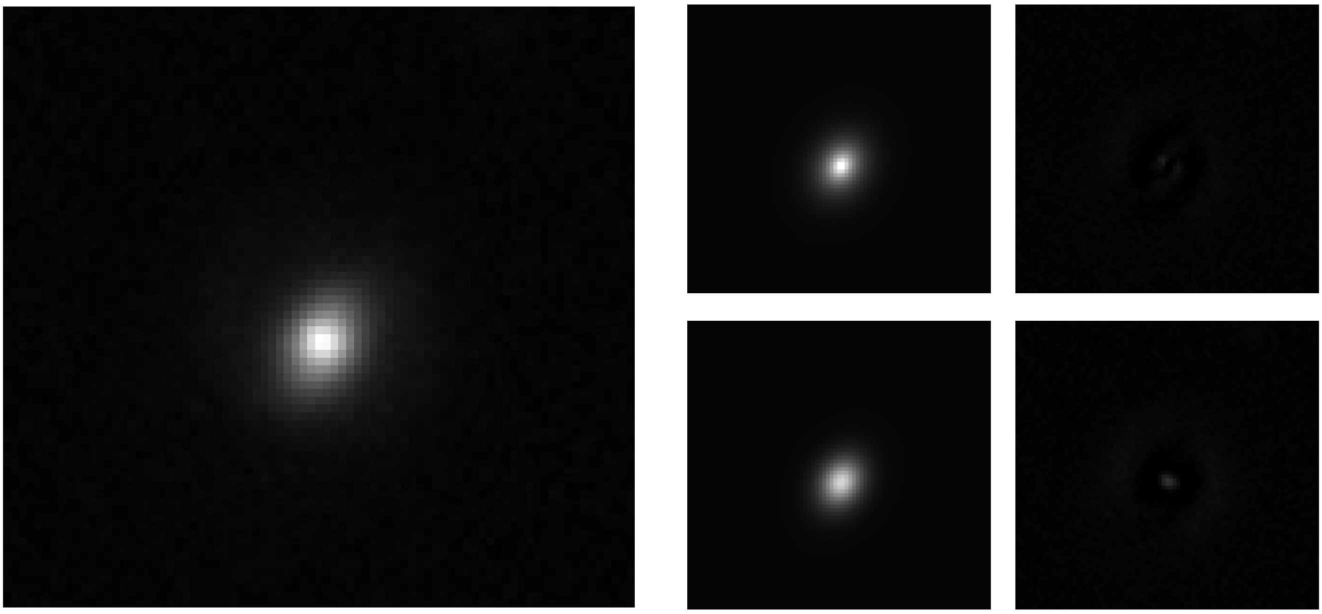
## 5.2 FRB 20180924B



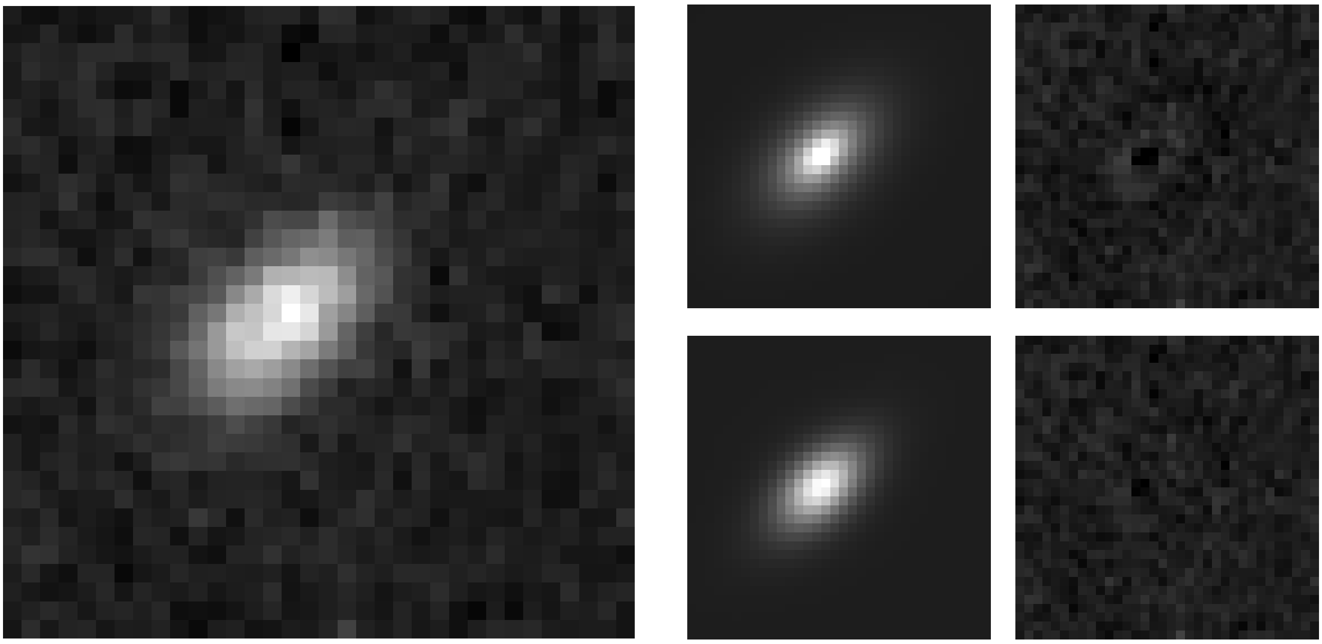
### 5.3 FRB 20190102C



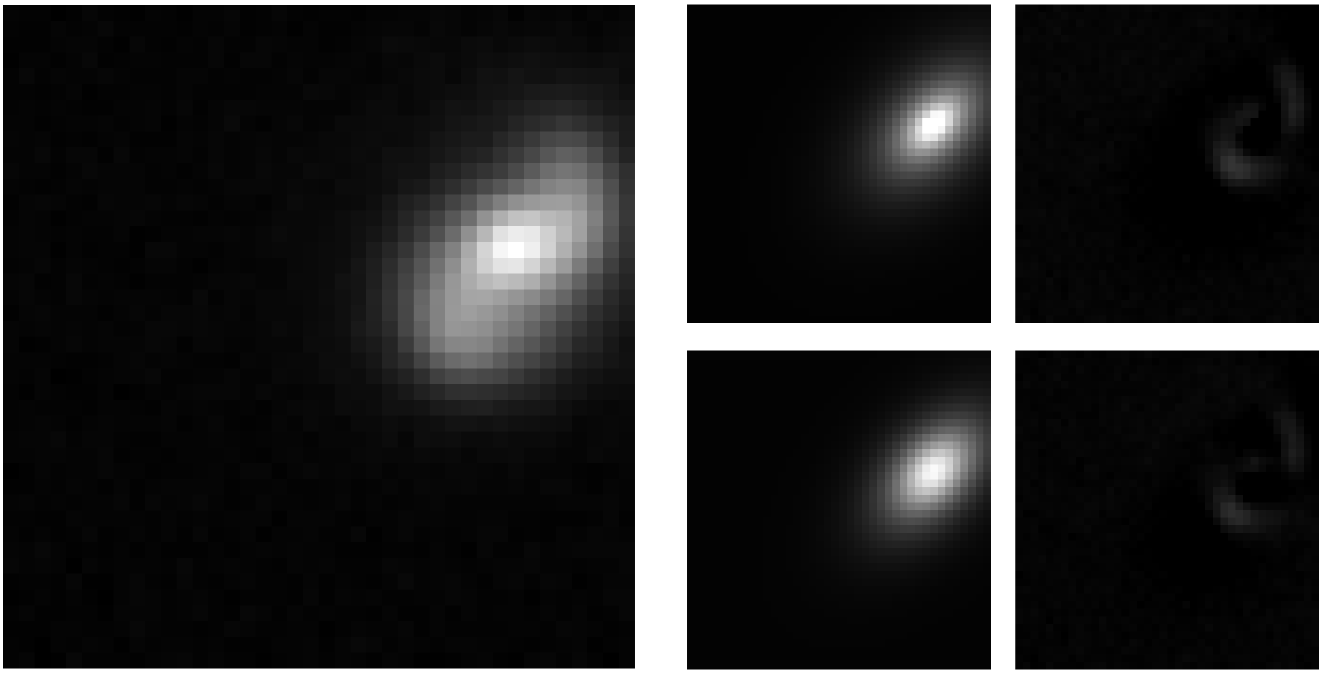
### 5.4 FRB 20190608B



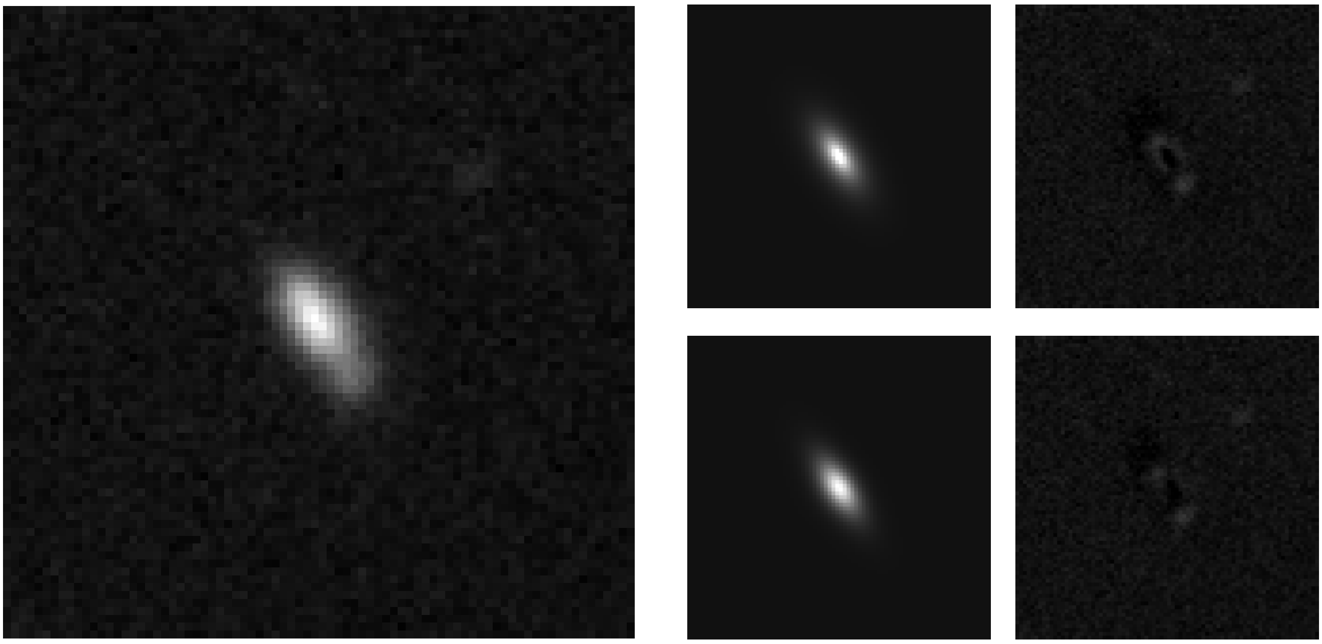
### 5.5 FRB 20190714A



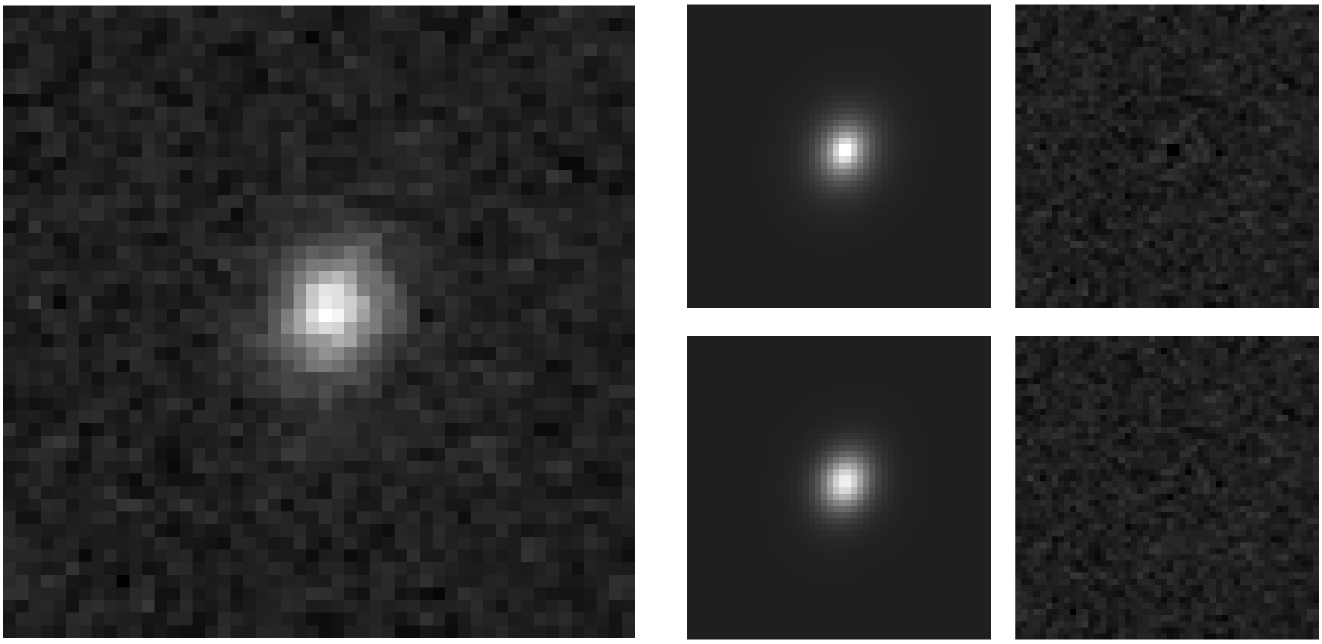
### 5.6 FRB 20191001A



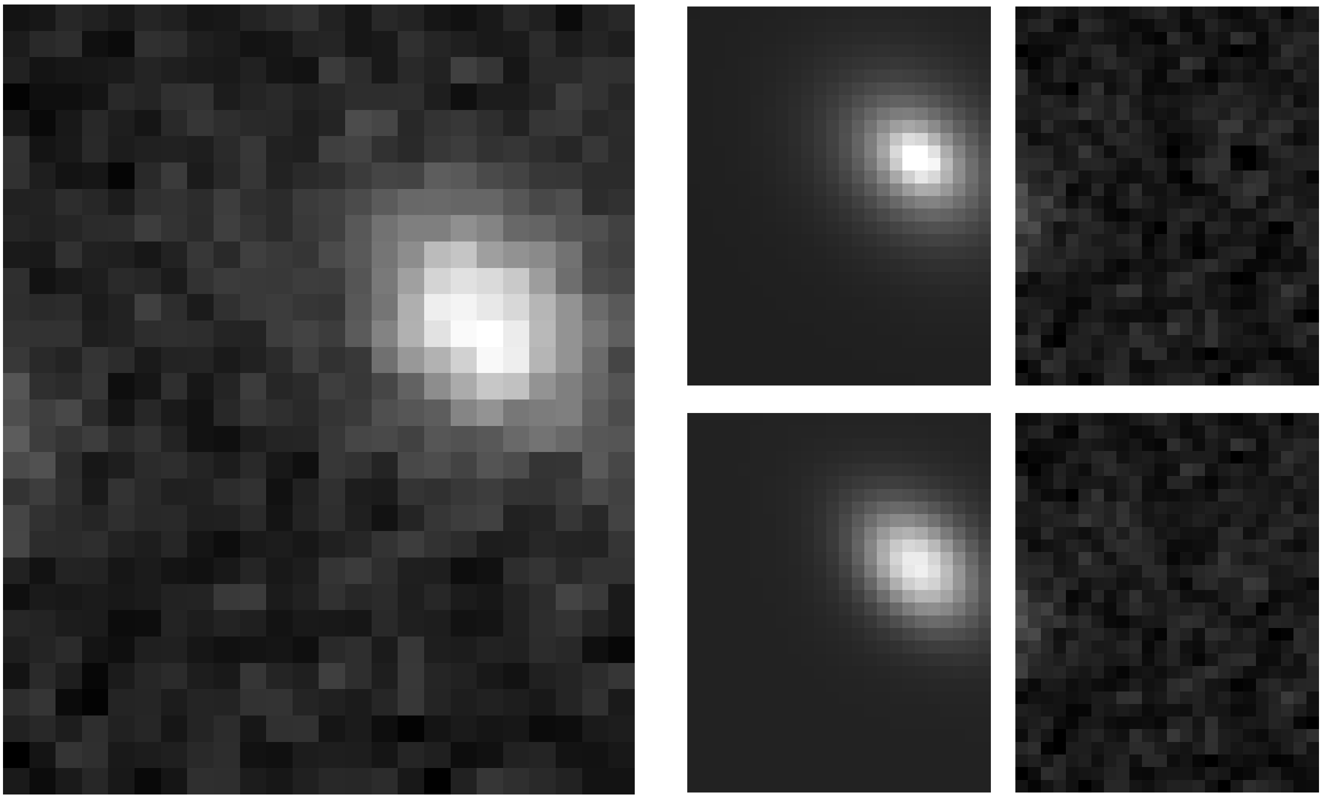
### 5.7 FRB 20200906A



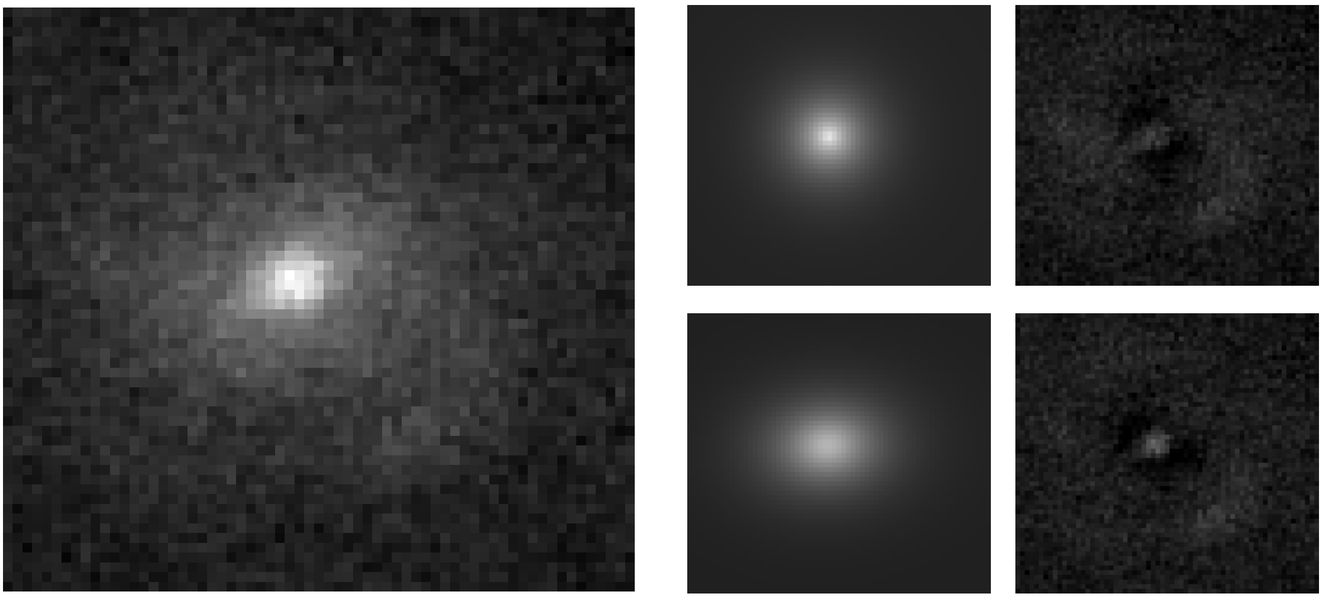
### 5.8 FRB 20210320C



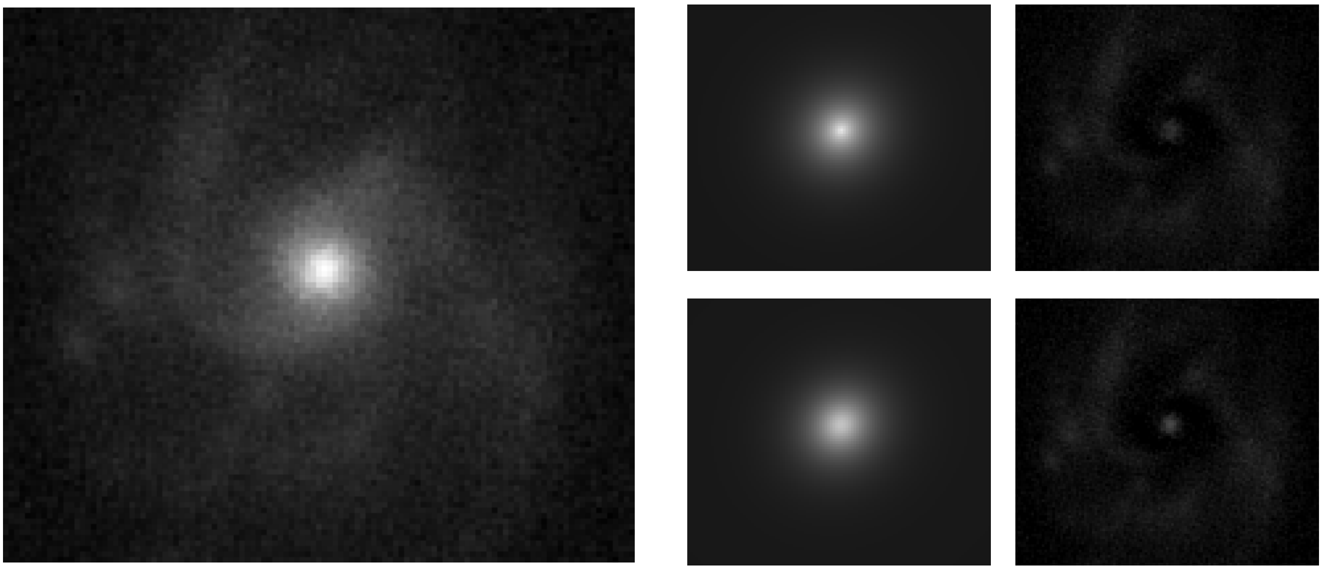
### 5.9 FRB 20210410D



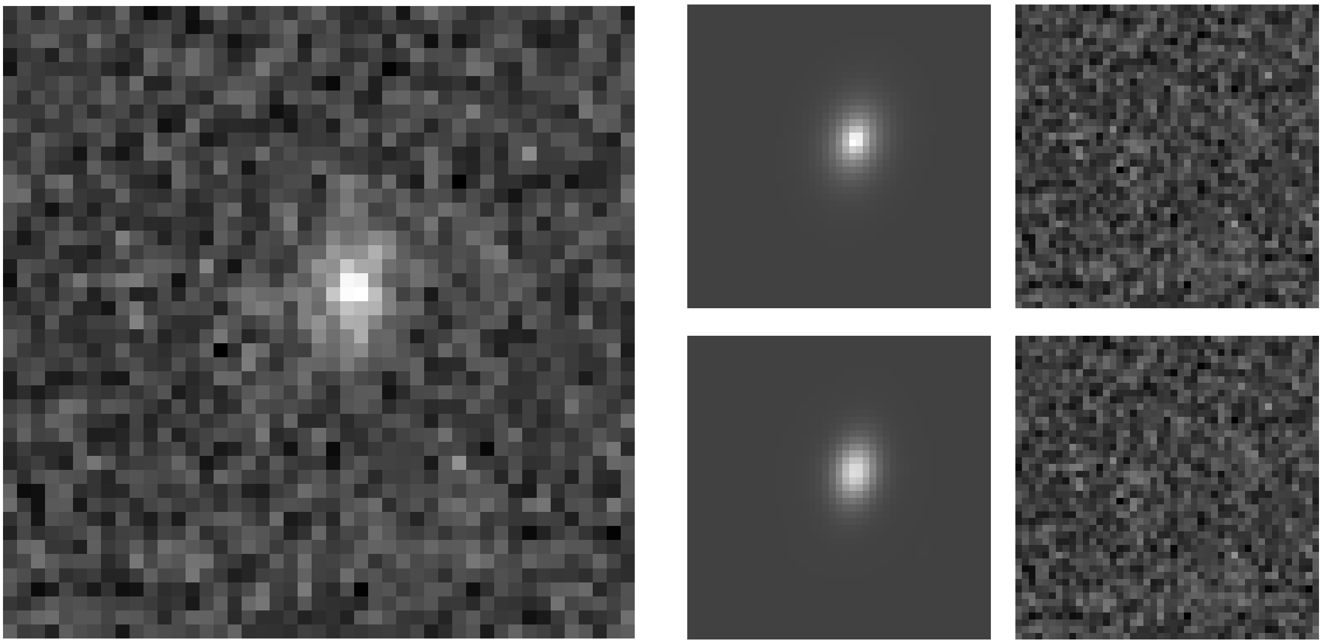
### 5.10 FRB 20210807D



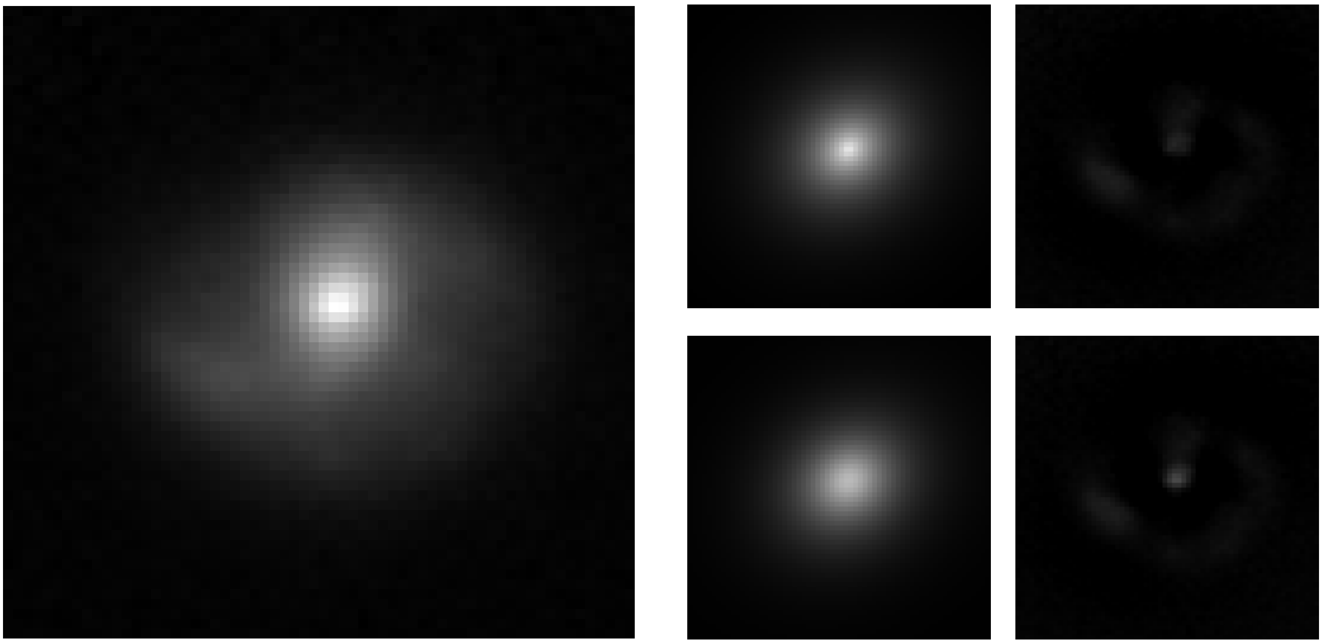
### 5.11 FRB 20211127I



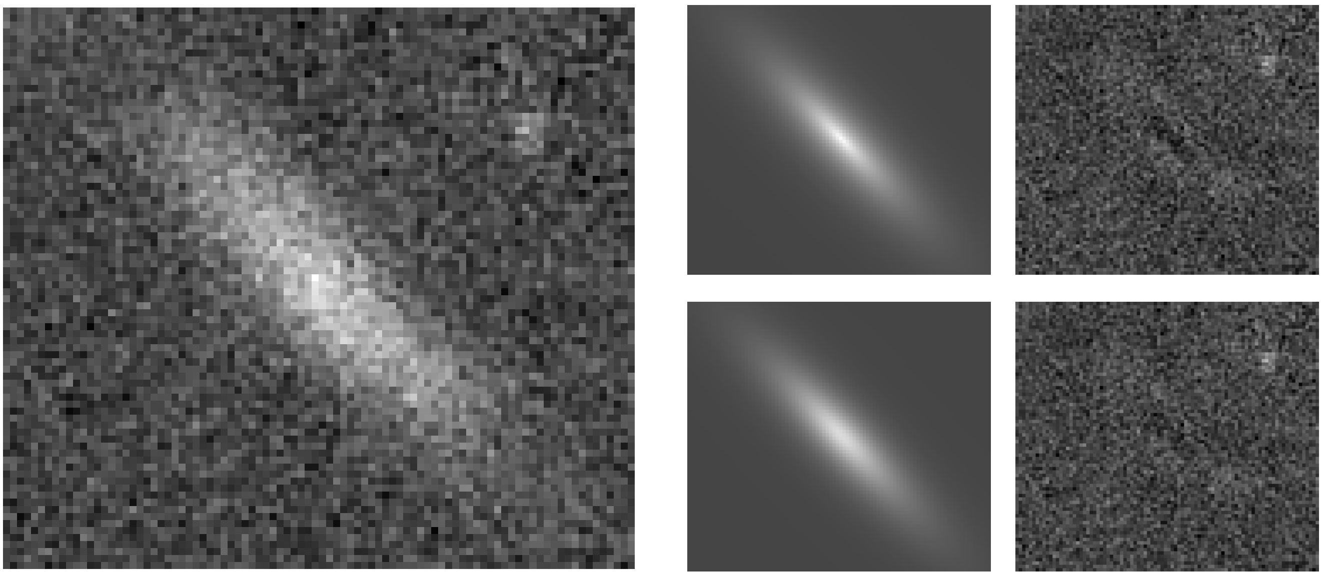
### 5.12 FRB 20211203C



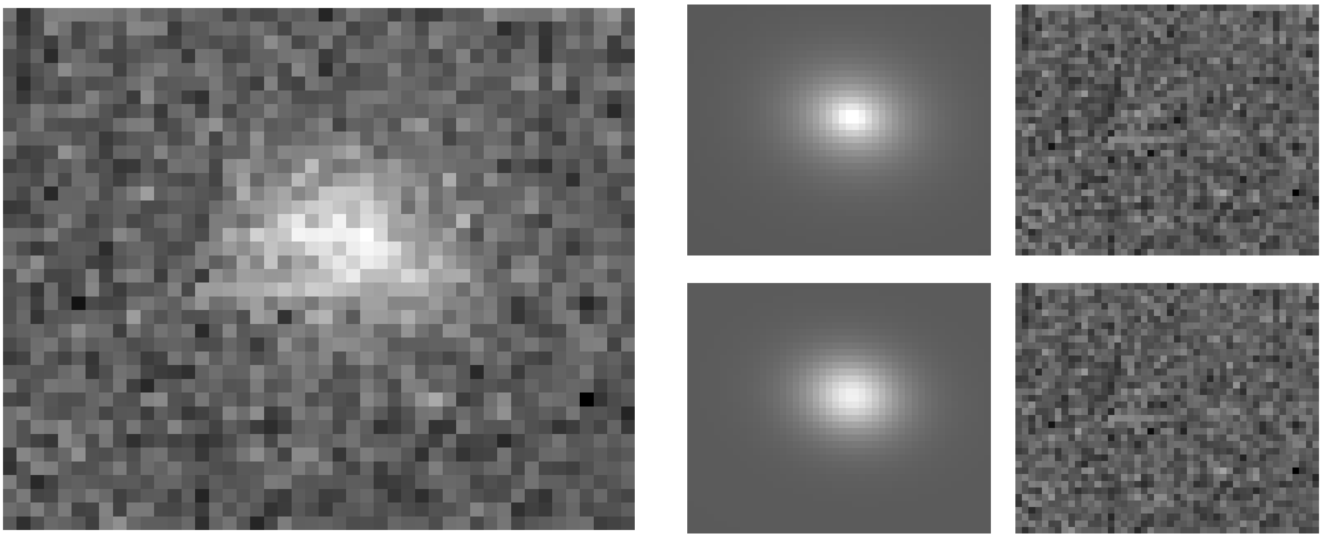
### 5.13 FRB 20211212A



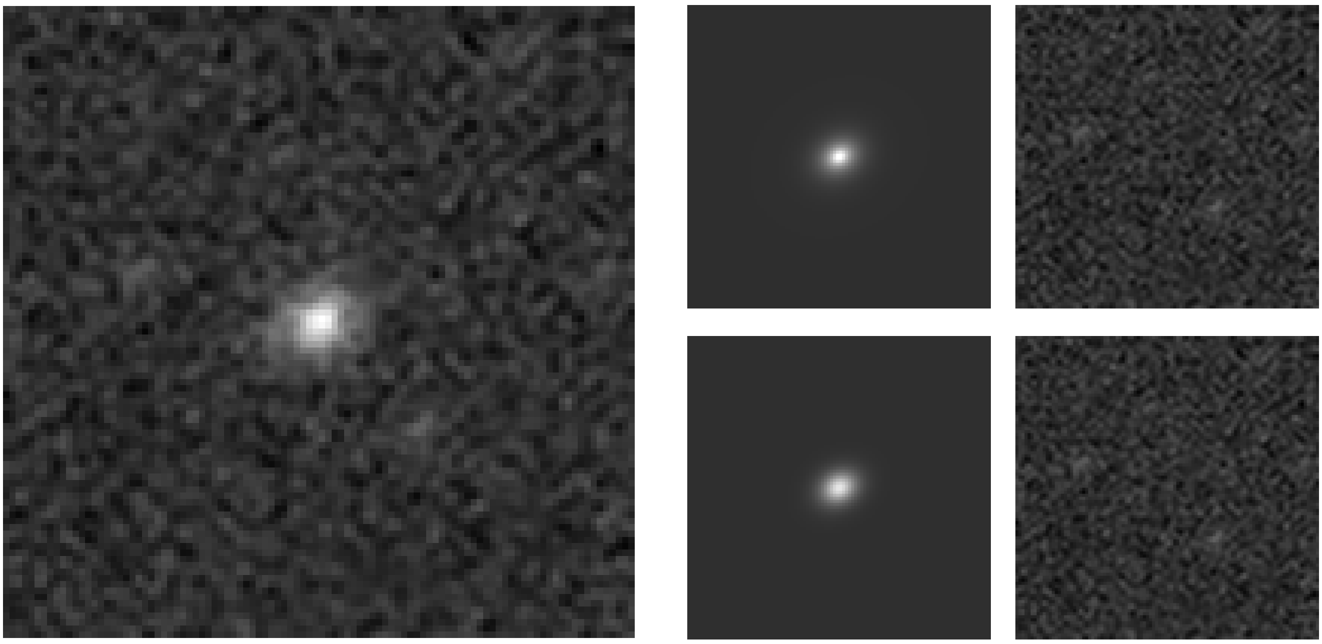
### 5.14 FRB 20220207C



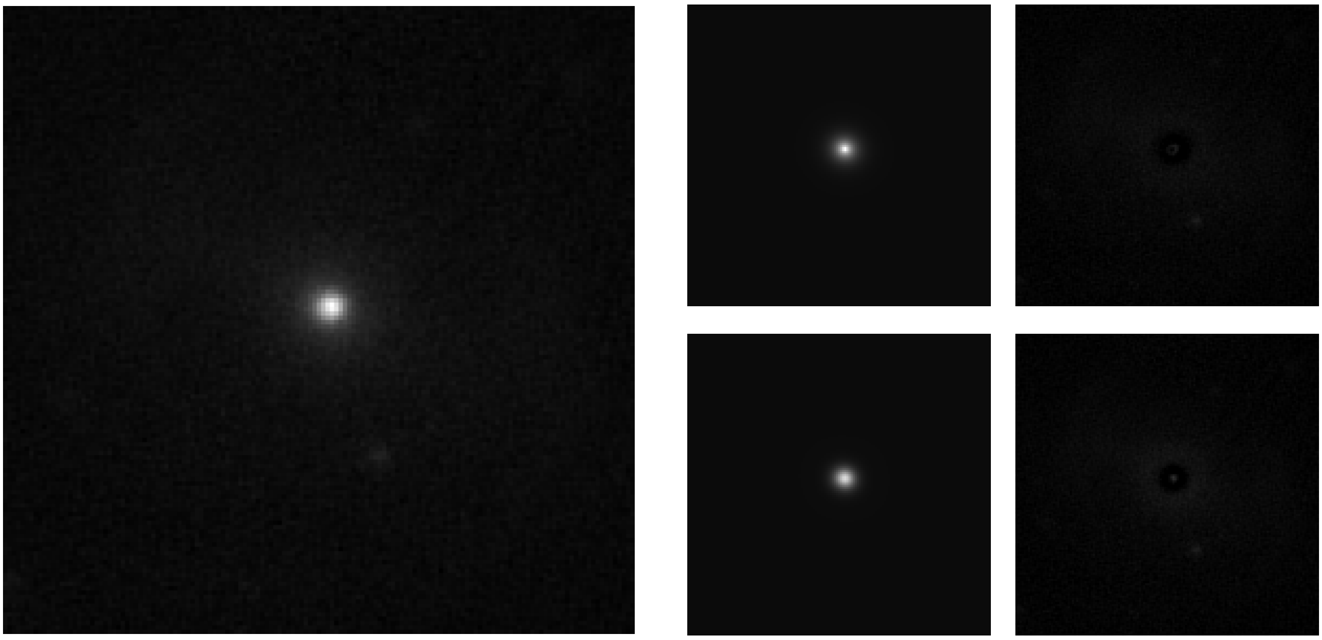
### 5.15 FRB 20220307B



### 5.16 FRB 20220310F

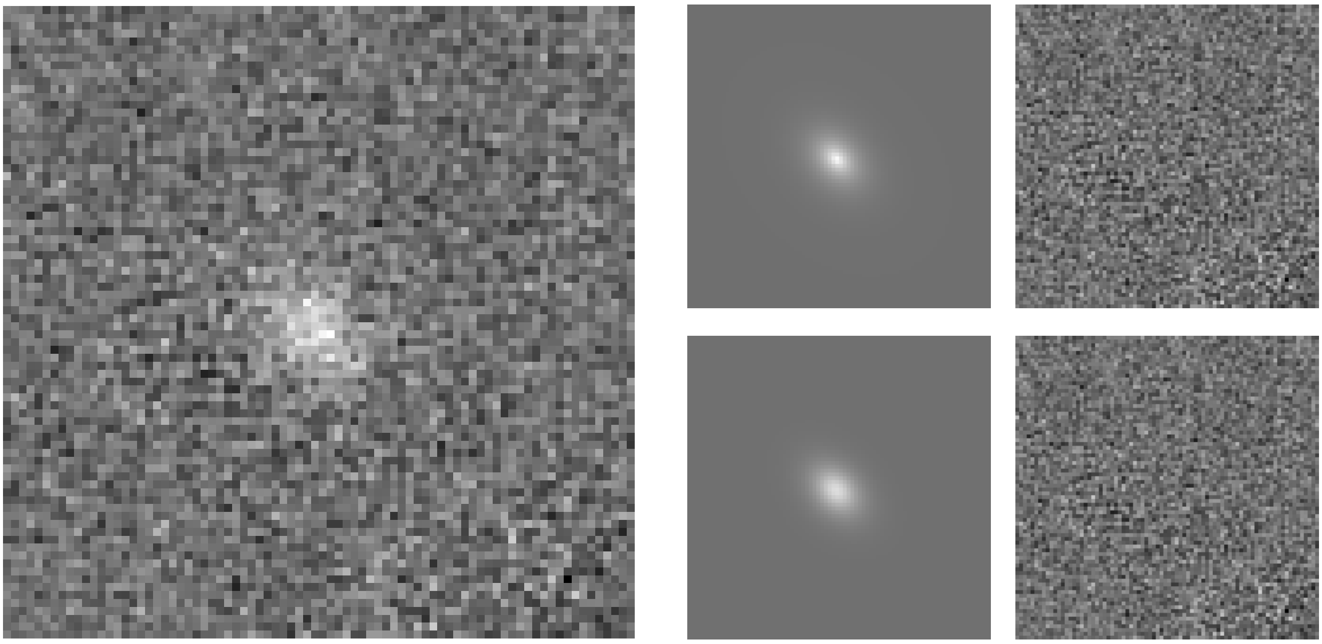


### 5.17 FRB 20220319D

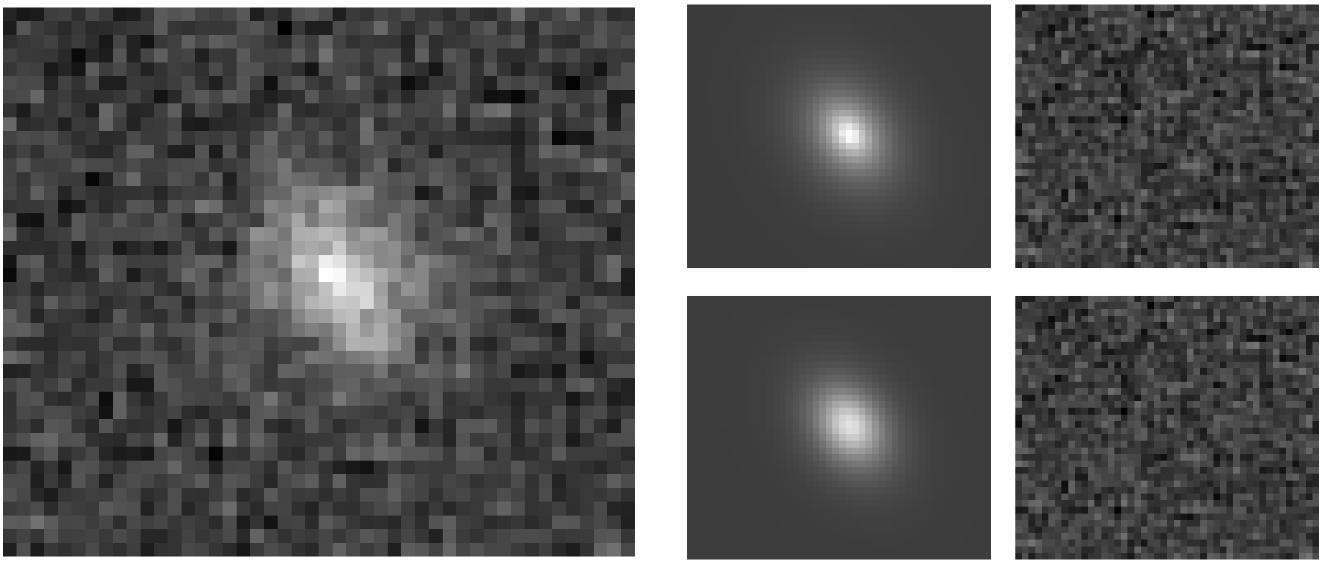


### 5.18 FRB 20220509G

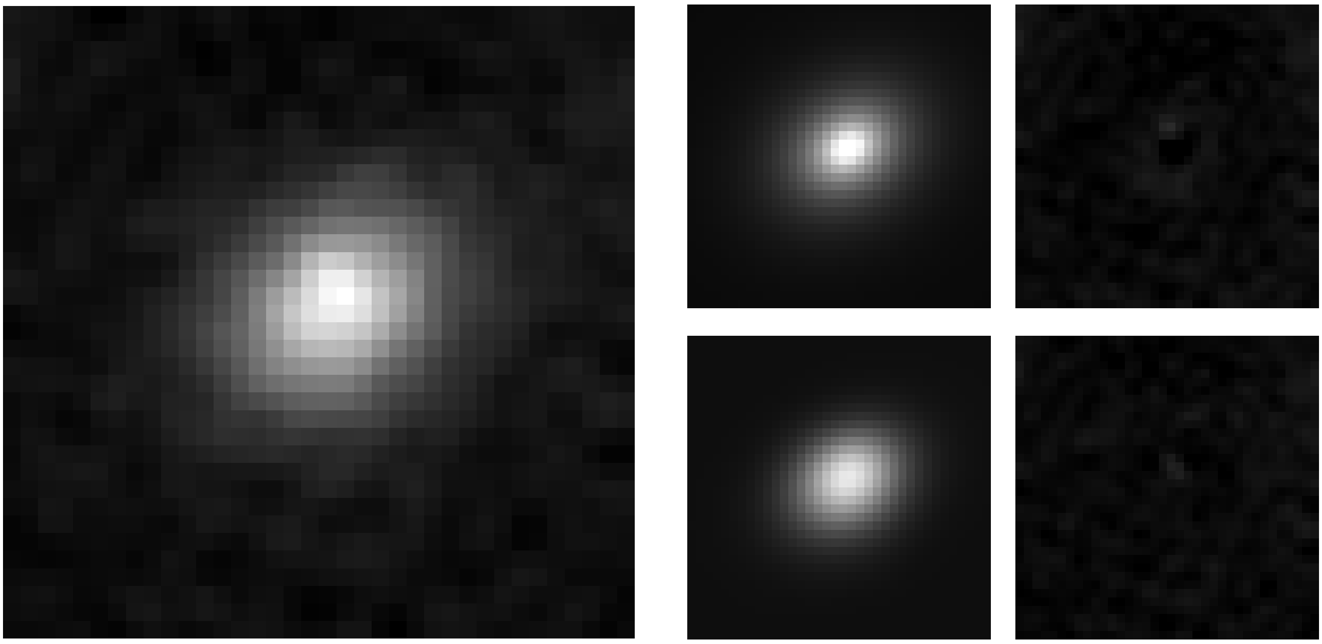
### 5.19 FRB 20220825A



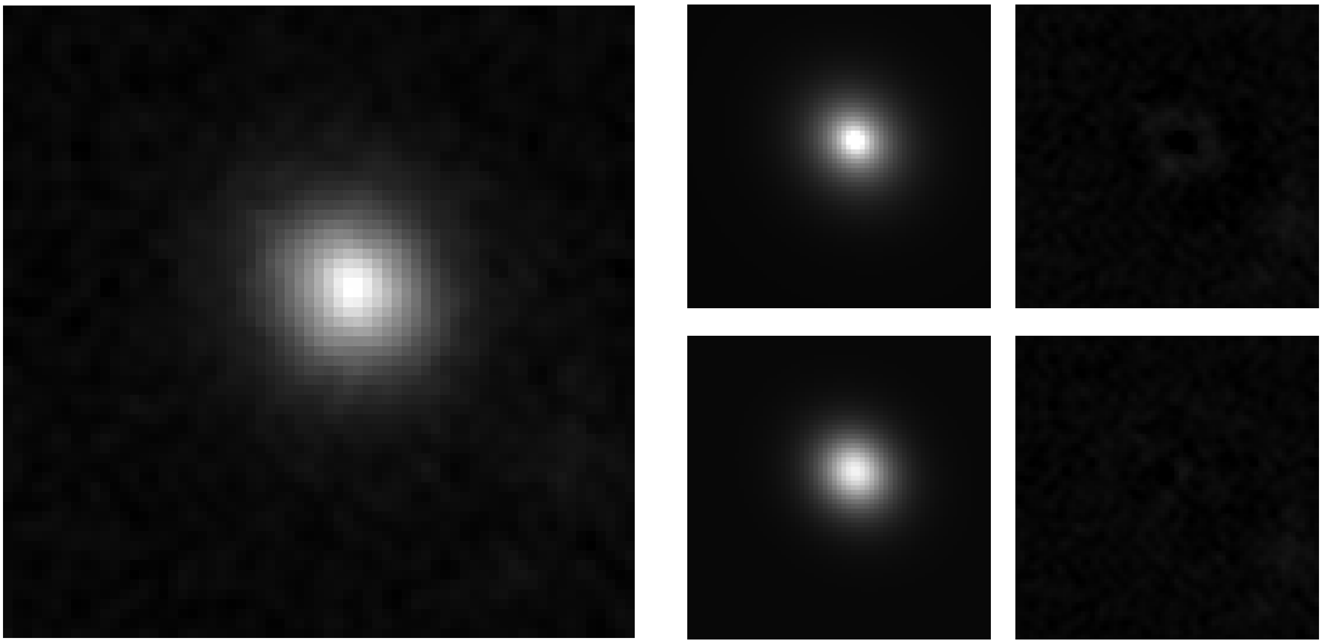
### 5.20 FRB 20220912A



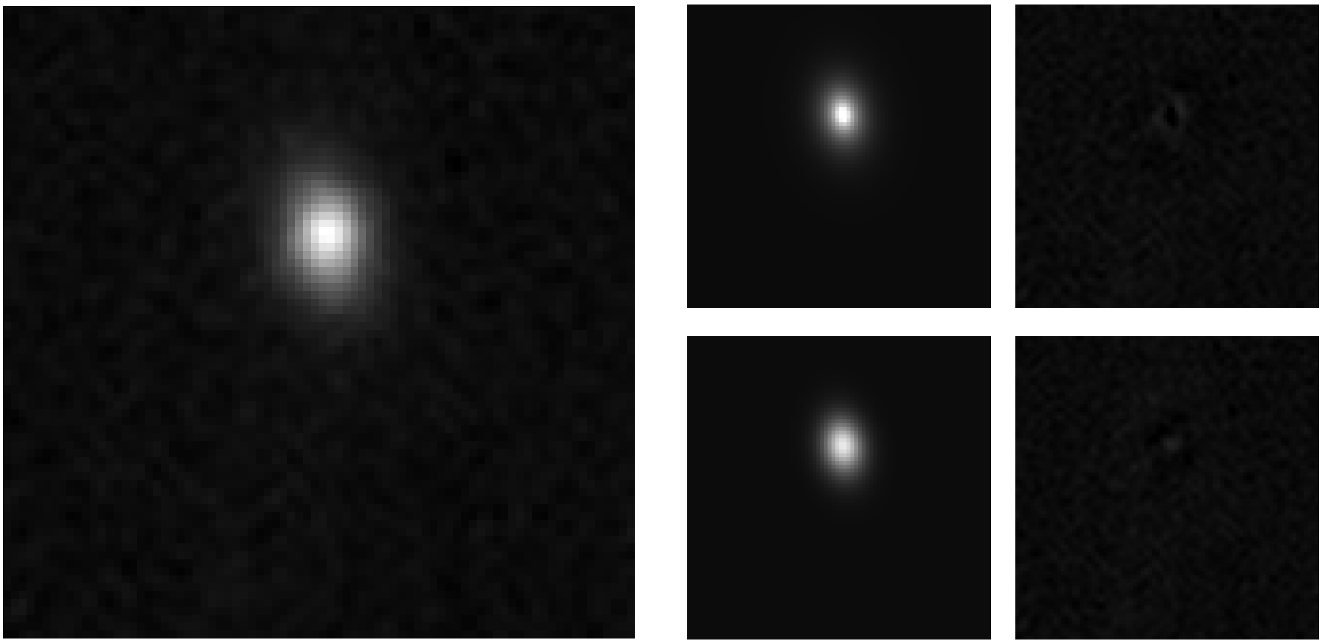
### 5.21 FRB 20220914A



### 5.22 FRB 20220920A



### 5.23 FRB 20221012A



## 6 Conclusion

All GALFIT image, model, and residual panels in this appendix use a common linear scale per FRB, where the image min/max are reused for both model and residual for consistent visual comparison.